# Railway Guages

In the first step, we have to import CSV file into a Pandas library. Pandas is used for handling and analyzing tabular data in the form of data-frame, flexible way to create data manipulation file which loads the dataset and also preview the data and also identifies the total value. Matplotlib is used to create visualizations like bar graphs and charts. Let us see some features.

In order to load the CSV file using pandas, we need to do following step.

.th

```
import pandas as pd
import matplotlib.pyplot as plt

# Load data
df = pd.read_csv('railway_gauges.csv')

# Display first 5 rows
print(df.head())

# Get row with maximum Total value (alternative method)
max_row = df[df['Total'] == df['Total'].max()]
print("\nRow with maximum Total:\n", max_row)
```

Plotting graphs is used for Matplotlib.pyplot.

The third line indicates a CSV file read and a dataframe named as df.
The dataset's first five rows will indicates the head() function. Whether it is loaded correctly or not. After the dataset is loaded it may get the initial. This section of code indicates the rows with the highest total value. the maximum value of total column is determined as df['Total'].max(). Only the row or rows with max value the condition, which may filters the DataFrame.

```
      Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891           464   6876
1  1965-66         3603         2913           470   6986
2  1966-67         3615         2911           481   7007
3  1967-68         3601         2920           506   7027
4  1968-69         3603         2921           508   7032

Row with maximum Total:
        Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
48  2012-13         6577          379           216   7172
```

Below are some of the essential Pandas functions and attributes.

- iloc [ ] — is primarily an integer based location search — iloc[3]

- loc[] — is primarily a label based location search — loc['Broad Gauge']

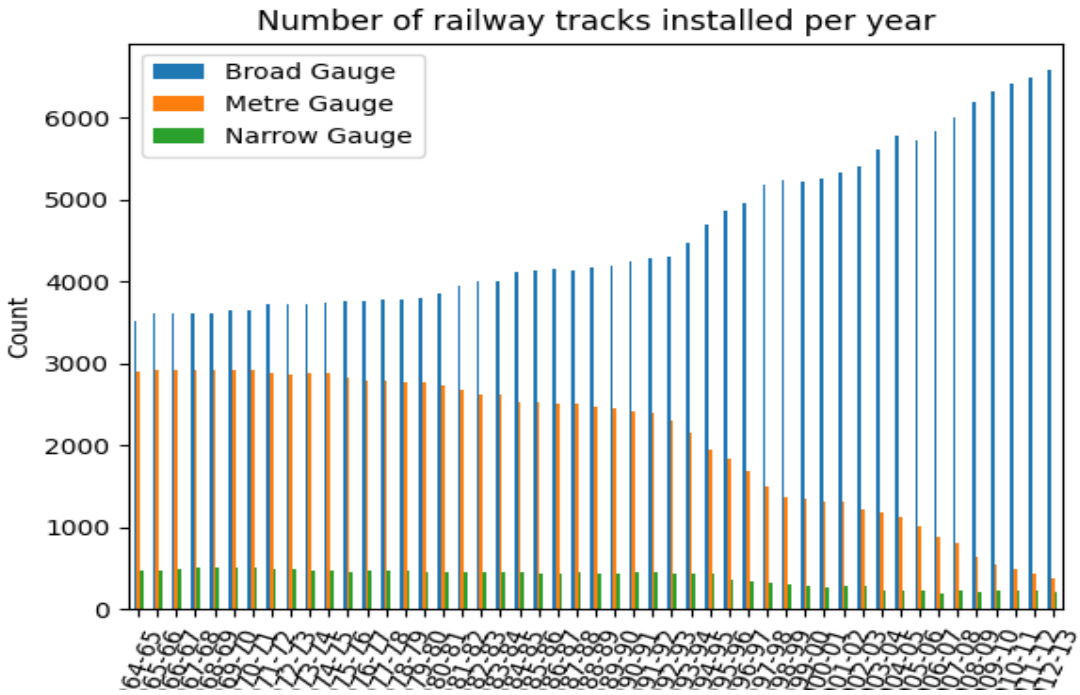
- idxmax()- is a function to identify the index of first occurrence of maximum over request axis

The remaining columns shown as values with year serving as the x-axis. The no.of railway gauges represented by each bar. The label on x-axis rotated by 70 degrees. Rotating improves readability and makes the chart look cleaner.
This is a common formatting step in data visualization.

Data Visualisation Using Matplotlib or Seaborn, we can plot the data flow in graphical representation format. In our example, below steps are used to see the data using bar chart.

```
# plotting data
df_plot = df.drop('Total', axis=1)
# Plot bar chart
df_plot.plot.bar(x='Year')

plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Count')
plt.title("Number of railway tracks installed per year")

plt.savefig('rail_gauges.png')
plt.show()
```

"Year" is written on the x-axis, and "Count" is written on the y-axis. The chart's meaning is explained the title. These can the graph's can be readability. The chart is shown in the screen . The plotted graph will displayed in the below.